



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Lecture 3: Elementary Probability Theory

2025/01/14

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Announcement

- Our exams will cover conceptual and computational questions.
- 80% of computational questions are selected from a subsequently published problem set, with only the numbers changed.

What to Expect Today

Outline:

- What is Probability?
- Key Probability Concepts: Random Experiment, Outcome, Sample Space, Events
- Axioms of Probability
- Examples and Applications

What is Probability? (Informal)

Real-World Example:

Think of the last time you made a decision under uncertainty: Should I bring an umbrella today? Should I invest in that stock?

Probability is the underlying math that helps us answer those kinds of questions!

Definition: Probability quantifies **how likely something** is to happen.

Why Study Probability?

It's the key to making better predictions about the uncertain world around us. We can estimate how likely it is that you'll get a 2 when rolling a die, or how likely you are to survive a game of "Russian Roulette" (without the bullet).

The Power of Probability – The Birthday Paradox

Challenge:

“What’s the probability that two people in a room of **100** share the same birthday?” You might be surprised at how high this probability is!

The Power of Probability – The Birthday Paradox

Hint: The answer isn't nearly as low as you think. The probability is actually much higher with 100 people—**it's over 99%!**

We will explore this in more detail as we define probability more rigorously.

Probability Theory

- A mathematical framework to model and quantify the uncertainty in real world.
- Example: I randomly select a student and ask for her/his birthday. I am not sure about her/his answer.
- Probability theory can help quantify the likelihood of each answer.

Consider a more complex situation

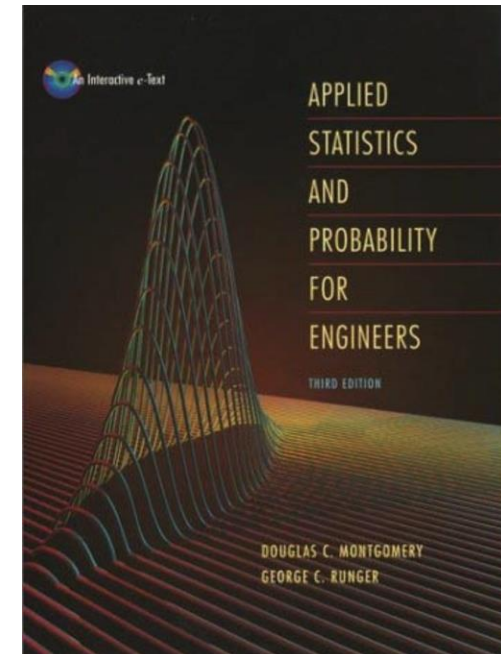
- Many students registered for this class
- What is the likelihood/probability that at least two students have the same birthday?

Probability and Statistics

- The basic problem that we study in probability is:
Given a data generating process, what are the properties of the outcomes?
- The basic problem of statistical inference is the inverse of probability:
Given the outcomes, what can we say about the process that generated the data?

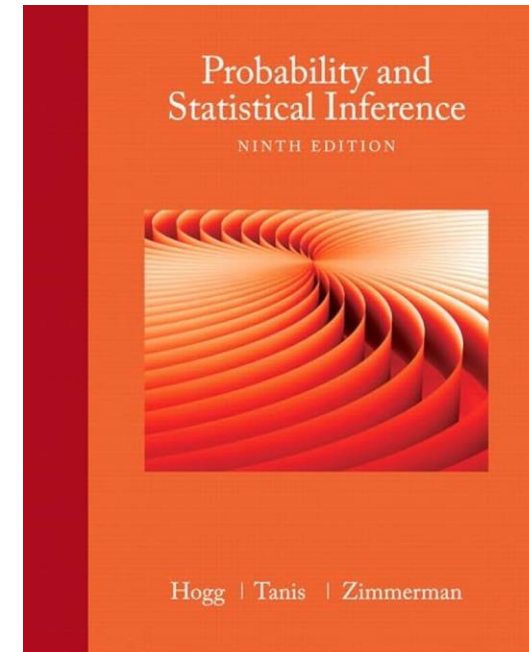
Suggested reading material:

- Applied Statistics and Probability for Engineers, Third Edition, Douglas C. Montgomery and George C. Runger.
- Chapter 2.1 – 2.3



Suggested reading material:

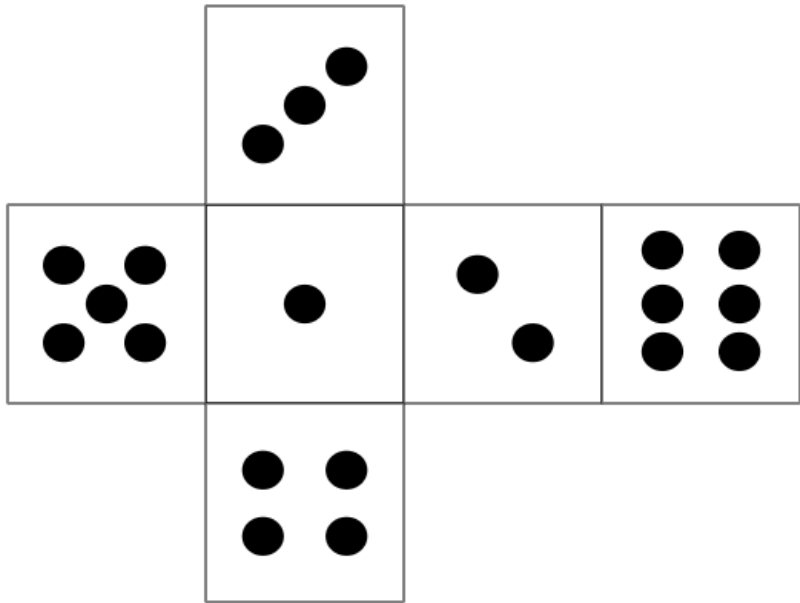
- Probability and Statistical Inference, 9th edition, Hogg, McKean and Craig.
- Chapter 1



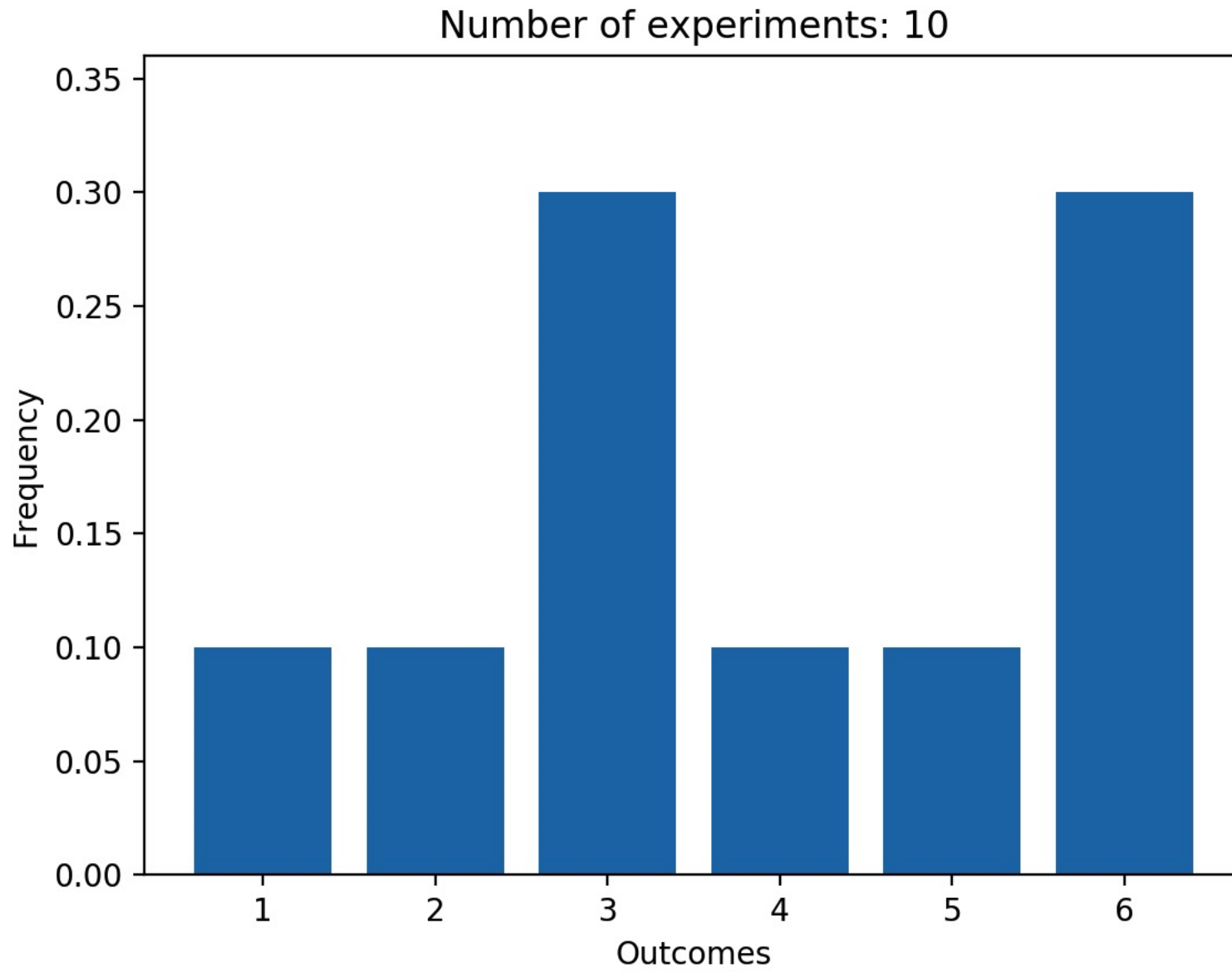
1. Definition of Probability

Given model, predict outcome.

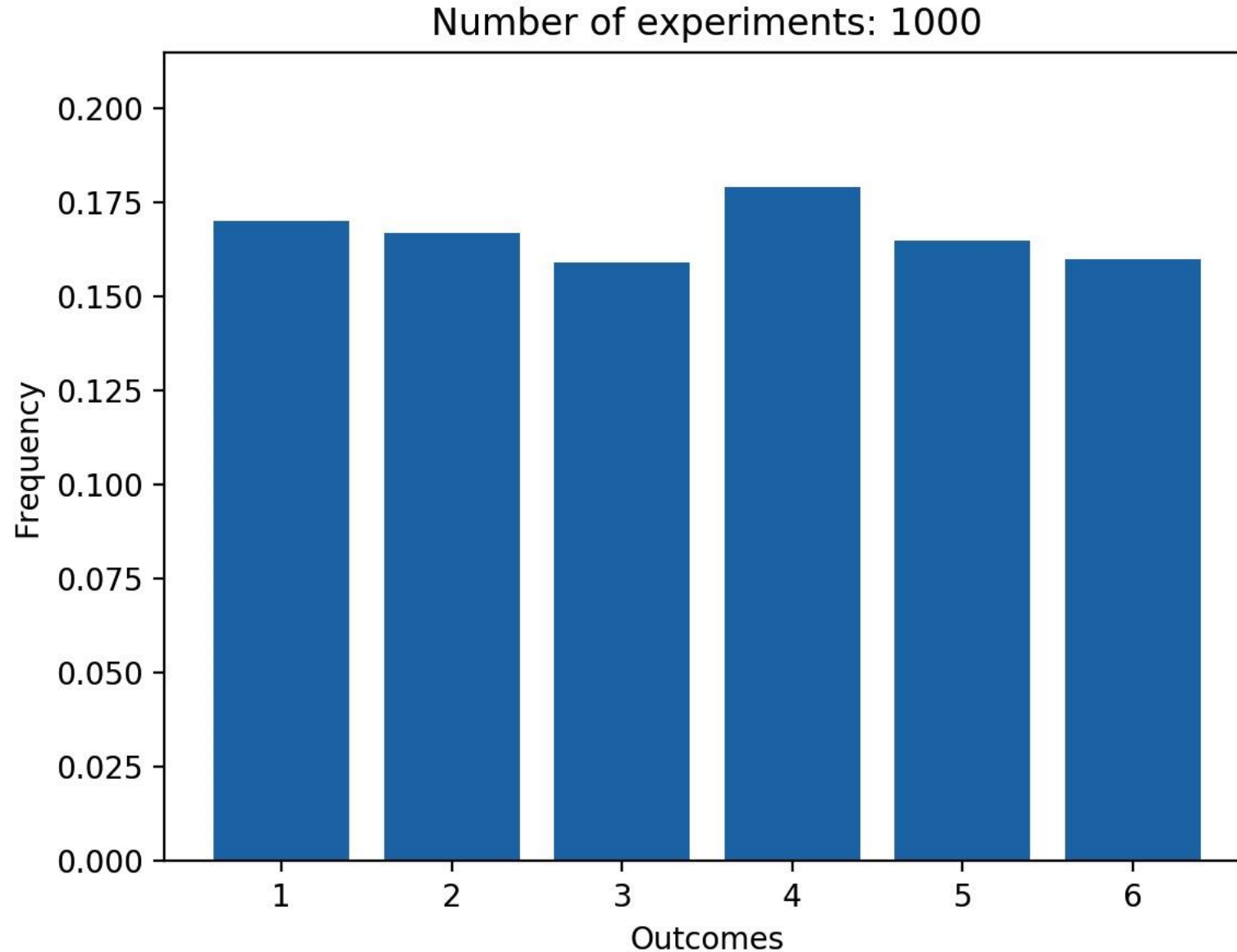
- Suppose you roll a die 10 times. How many times do you expect to roll a 2? Make your best guess.



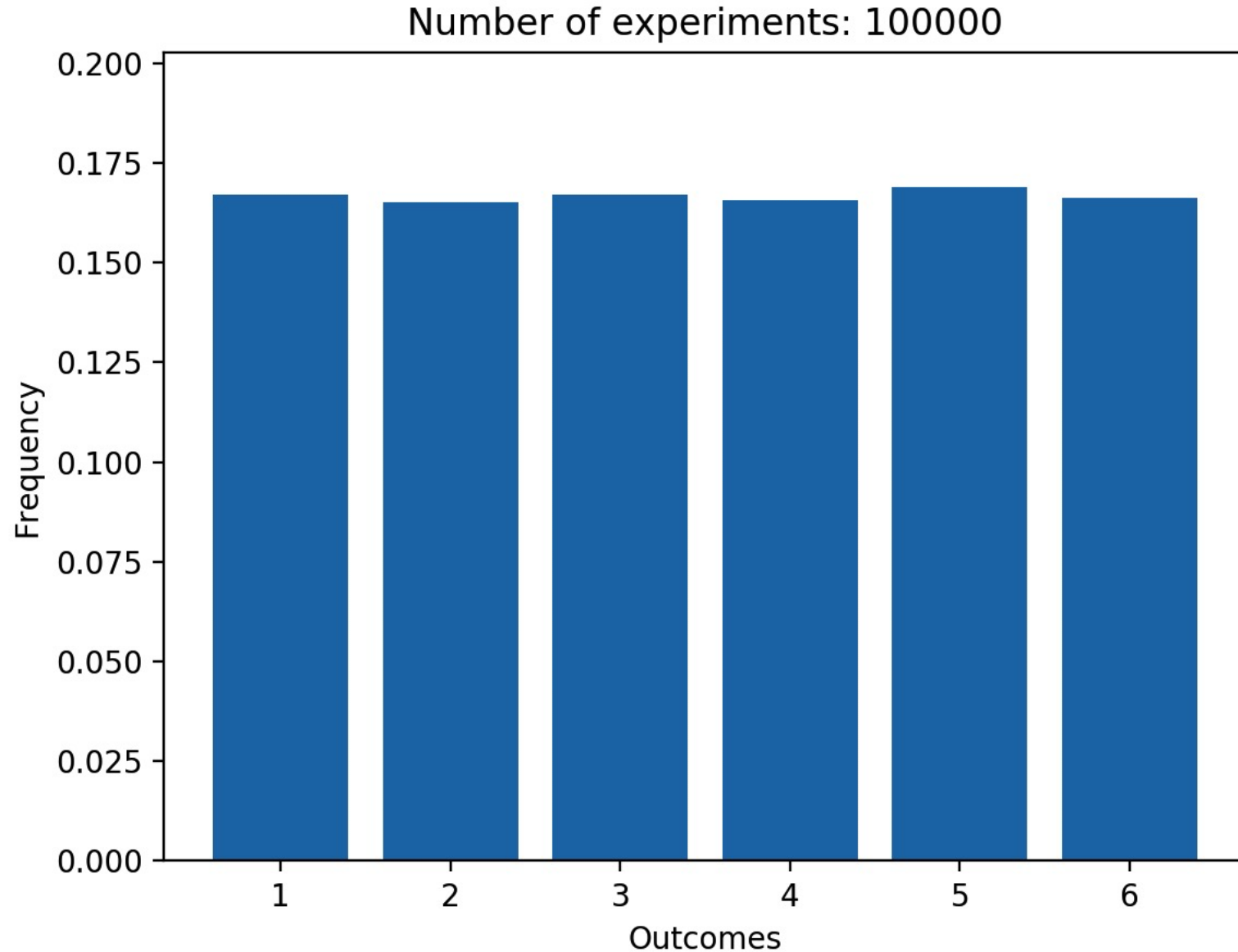
Roll a fair dice 10 times



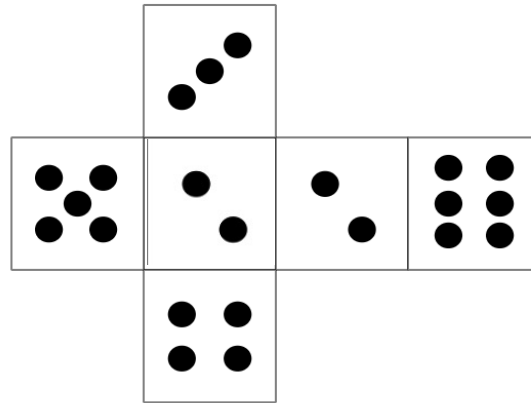
Roll a fair dice 1000 times



Roll a fair dice 100000 times



- Experiment: roll an unfair die 600000 times.



- How many times do you expect to roll a 2?

What do they have in common?

- We are doing an experiment that can result in different outcomes.
- If repeating the experiment infinitely many times, the **frequency** of getting a certain outcome will converge to a certain number.

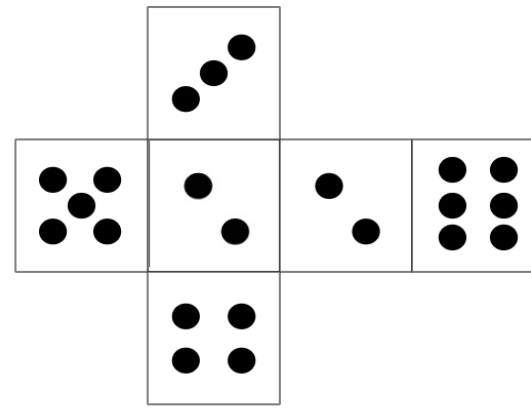
What is Probability?

- An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**.
- Probability is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur.

What is Probability?

- Random Experiment:
- Consider one possible outcome: ω
- The outcome ω happens with probability $P(\omega)$
- **It means:**
 - If we repeat this experiment N times
 - We observe n occurrences of the outcome ω .
 - Then if N goes to infinity, n/N will approach $P(\omega)$.

- Experiment: roll a fair die once
- All possible outcomes: $1,2,3,4,5,6$.
- What is the probability of obtaining the outcome i , denoted by p_i , for $i = 1,2,3,4,5,6$
- When you roll the die N times and observe the outcome i a total of N_i , as N goes to infinity, N_i/N will approach p_i .
- When the die is fair, N_i/N approaches $1/6$ and thus $p_i = 1/6$



- Experiment: roll an unfair die
- What is the probability that the outcome is 2.
- You roll the die N times and observe the outcome 2 a total of N_2 times; as N goes to infinity, N_2/N will approach p_2 .
- For this die, N_i/N approaches $1/3$ and thus $p_2 = 1/3$

Questions

- Can a probability be negative?
- Can a probability be larger than 1?
- When rolling a fair die, what is the probability of getting an outcome > 3 ?

Questions

- Can a probability be negative? No
- Can a probability be larger than 1? No
- When rolling a fair die, what is the probability of getting an outcome > 3 ?

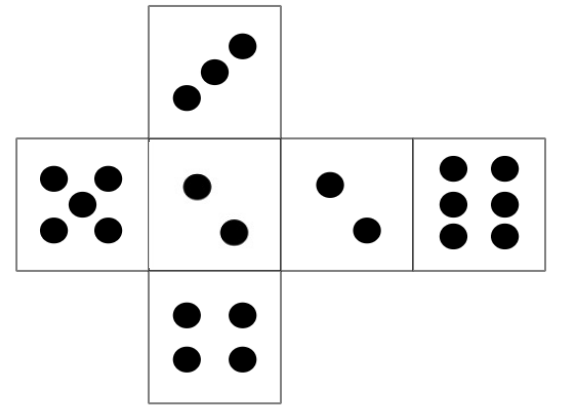
Terminologies

- **Random Experiment:** a repeatable procedure
- **Sample space:** set of all possible outcomes Ω .
- **Event:** a subset of the sample space.
- **Probability function, $P(\omega)$:** gives the probability for each outcome $\omega \in \Omega$
 - Probability is between **0** and **1**
 - Total probability of all possible outcomes is **1**.
 - If $A = \{\omega_1, \omega_2, \omega_3, \dots\}$, $P(A) = P(\omega_1) + P(\omega_2) + P(\omega_3) + \dots$

Example 1

- Experiment: roll a fair die, report the number
- Sample space: $S = \{1,2,3,4,5,6\}$.
 - An outcome is a single result of a random experiment, incorporating the **full** information of your observation.
 - You can use letters, numbers, or other symbols to represent one outcome.
- Probability: $P(\{i\}) = P(i) = 1/6, i = 1,2,3,4,5,6$.
 - We delete $\{\}$, if it contains a single outcome.
- Event A that the number is > 3 : $\{4,5,6\}$
 - $P(A) = P(4) + P(5) + P(6) = 3/6 = 1/2$

Example 2



- Experiment: roll an unfair die, report the number
- Sample space: S
- Probability: $P(i)$ for $i \in S$.
- Event A: the number is < 4 :
 - $P(A)$?

Properties

- Axioms of Probability
 - ✓ $P(S) = 1$, S is the sample space
 - ✓ $0 \leq P(A) \leq 1$ for any event A
 - ✓ For any two events A and B , if they do not contain any common outcome, we have

$$P(A \cup B) = P(A) + P(B)$$

Example 3

The outcome of an experiment X is always either 1 or 2.

Suppose that $P(X = 1) = 1/4$. What is the value of $P(X = 2)$?

2. Some Remarks

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Sample Space

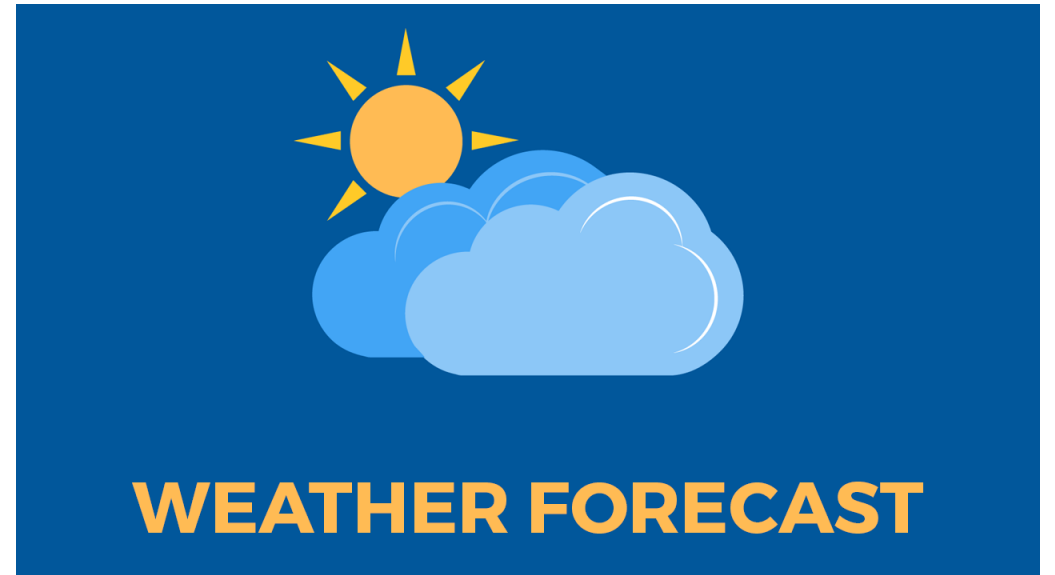
Example 4

- You bought 3 blind boxes which contain either Lord Voldemort or Harry Potter

Describe the set of possible outcomes.

Example 5

- What will be the highest temperature tomorrow?



What's the difference?

- Discrete or continuous: countable (listable) or not?

A sample space is **discrete** if it consists of a finite or countable infinite set of outcomes.
A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.

Exercise

- Which of the following are continuous?
 1. The sum of numbers on a pair of two dice.
 2. The possible sets of outcomes from flipping ten coins.
 3. The possible sets of outcomes from flipping (countably) infinite coins.
 4. The possible values of the temperature outside on any given day.
 5. The possible times that a person arrives at a restaurant.

For 5:

Continuous:

- If we assume a person can arrive at any moment, down to fractions of a second, within a given interval (e.g., 9:00:00 AM to 8:59:59 PM), the arrival time is treated as a **continuous variable**.

Discrete:

- If we consider arrival times in specific increments, such as minutes, hours, or other fixed intervals, then the arrival time becomes a **discrete variable**.

2. Some Remarks

Event

Events

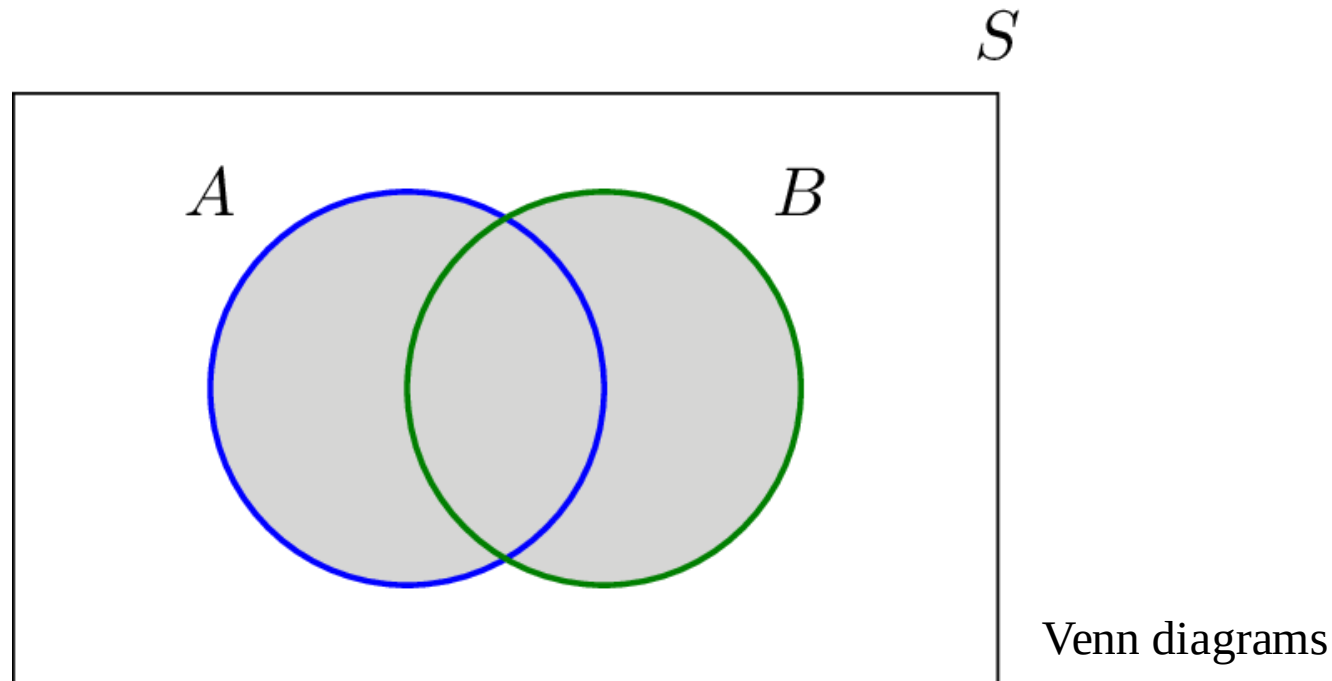
- Events are sets:
 - ✓ Can describe in words
 - ✓ Can describe in notation
- Experiment: toss a coin **2** times.
- Event -- You get **1** or more heads
 $= \{HH, HT, TH\}$

Set operations

- Events are sets, so we can use set operations
 - ✓ Unions
 - ✓ Intersections
 - ✓ Complements

Set operations - union

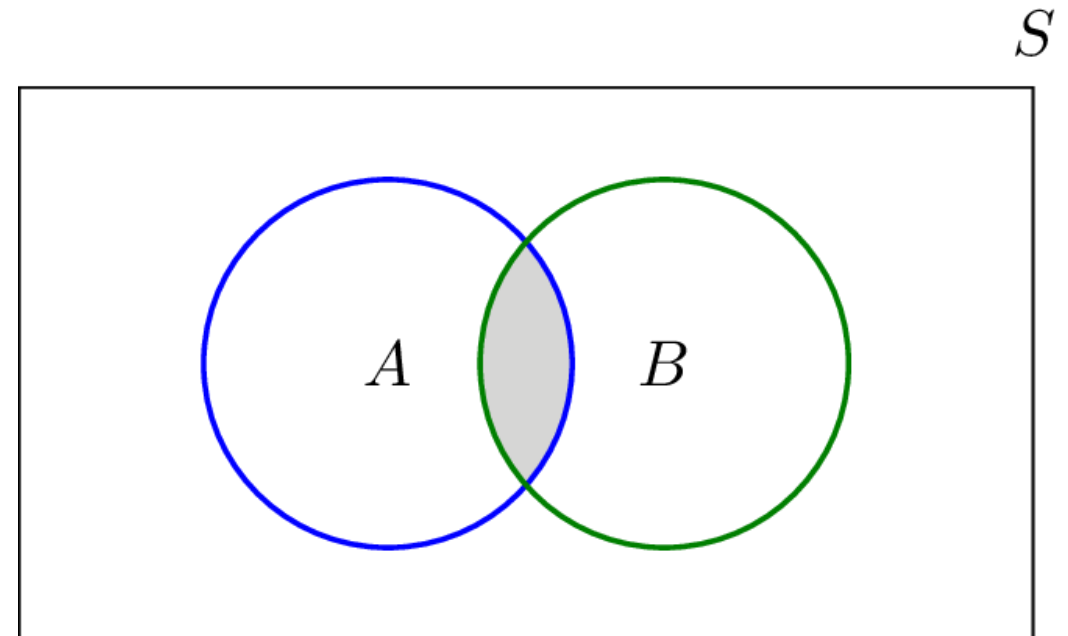
- The **union** of two events A and B is the event that consists of all outcomes that are contained in either A or B .
- We denote the union as $(A \text{ or } B)$ in words, and $A \cup B$ in notation



Set operations - intersection

- The **intersection** of two events A and B is the event that consists of all outcomes that are contained in both A and B .
- We denote the intersection as $(A \text{ and } B)$ in words, and $A \cap B$ in notation
- Two events A and B , such that $A \cap B = \emptyset$ are said to be **mutually exclusive**.

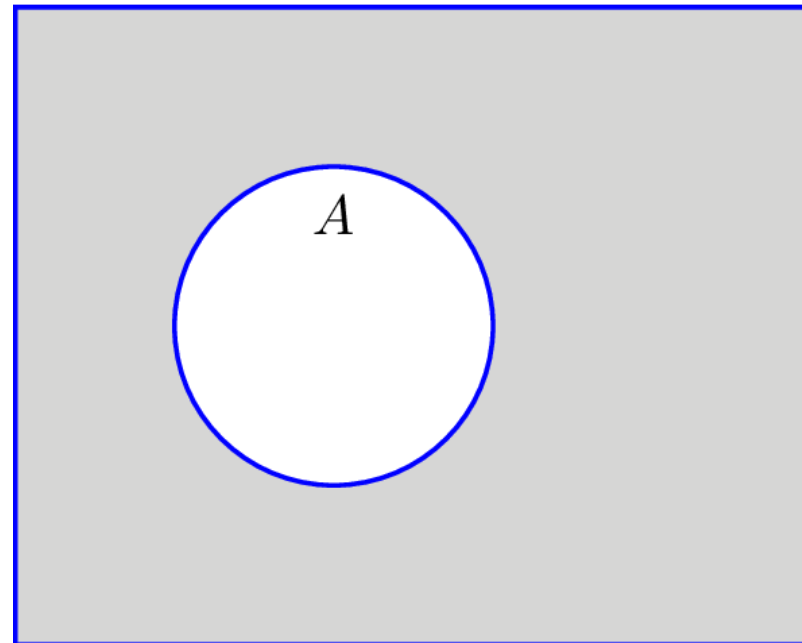
Note: The intersection can be an empty set.



Set operations - complement

- The **complement** of an events A in a sample space S is the set of outcomes in the sample space that are not in the event A .
- We denote the complement as (not A) in words, and A' or A^c in notation

Note: The complement can also be an empty set, it can also be the whole sample space.



Example 6

- Toss a coin 2 times: sample space: $\{TT, HT, TH, HH\}$
- Event A : there is at least one Head: $A = \{HH, HT, TH\}$
- Event B : there is at least one Tail: $B = \{TT, HT, TH\}$
- Question:
 - What is the event A or B , namely $(A \cup B)$?
 - What is the event A and B , namely $(A \cap B)$?
 - What is the complement of event A ?

Some useful properties

- The complement of the complement of an event is itself

$$(E')' = E$$

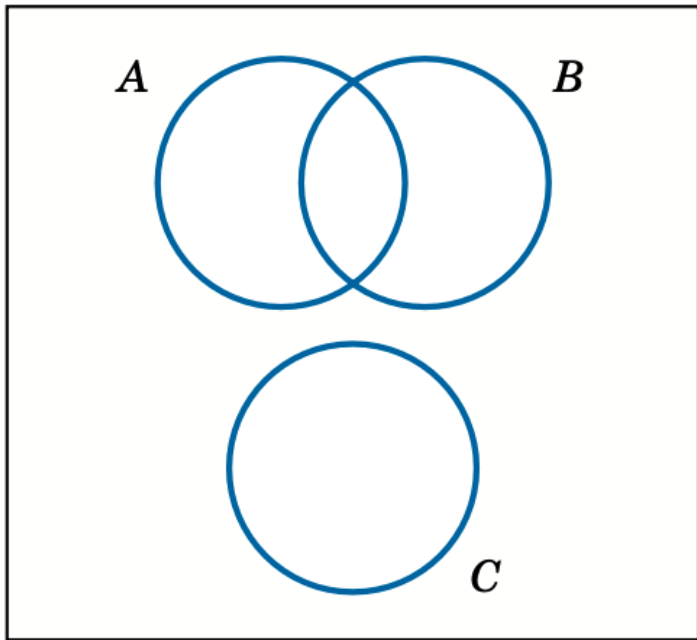
- The intersection and union does not depend on the order

$$A \cap B = B \cap A \quad \text{and} \quad A \cup B = B \cup A$$

- The complement of the union (intersection) is the intersection (union) of the complement

$$(A \cup B)' = A' \cap B' \quad \text{and} \quad (A \cap B)' = A' \cup B'$$

Exercise



Shade the region that corresponds to each of the following events:

(a) A'

(b) $(A \cap B) \cup (A \cap B')$

(c) $(A \cap B) \cup C$

(d) $(B \cup C)'$

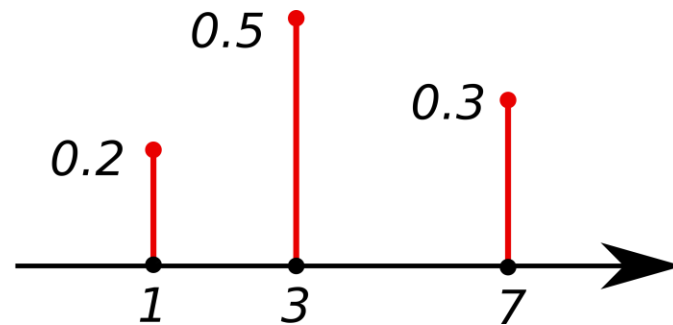
(e) $(A \cap B)' \cup C$

2. Some Remarks

Probability Function

Probability function

- Discrete:
 - ✓ **Probability mass function.**
 - ✓ $P(\omega)$: gives the probability for each outcome $\omega \in S$



Continuous case will be defined later.

Calculation of Event Probability?

By the definition...

- First, you write down $P(\omega)$ for each $\omega \in S$.
- Then, if $A = \{\omega_1, \omega_2, \omega_3, \dots\}$, $P(A) = P(\omega_1) + P(\omega_2) + P(\omega_3) + \dots$

Calculation of Event Probability?

By the definition...

- First, you write down $P(\omega)$ for each $\omega \in S$.

Tables can really help in complicated examples!

- Then, if $A = \{\omega_1, \omega_2, \omega_3, \dots\}$, $P(A) = P(\omega_1) + P(\omega_2) + P(\omega_3) + \dots$

Example 7

A random experiment can result in one of the outcomes $\{a, b, c, d\}$ with probabilities 0.1, 0.3, 0.5, and 0.1, respectively. Let A denote the event $\{a, b\}$, B the event $\{b, c, d\}$, and C the event $\{d\}$. Then,

Event	a	b	c	d
Probability	0.1	0.3	0.5	.1

Example 7

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Event	a	b	c	d
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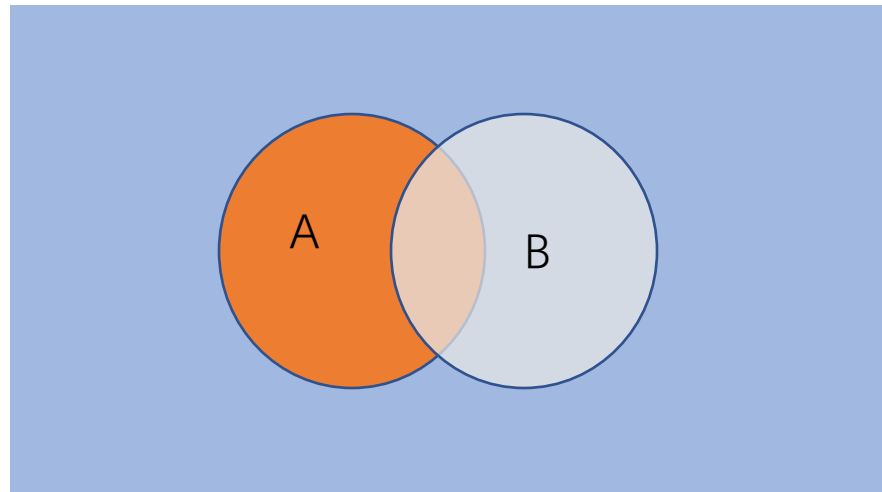
$$P(A) = 0.1 + 0.3 = 0.4$$

$$P(B) = 0.3 + 0.5 + 0.1 = 0.9$$

$$P(C) = 0.1$$

Calculation of Event Probability?

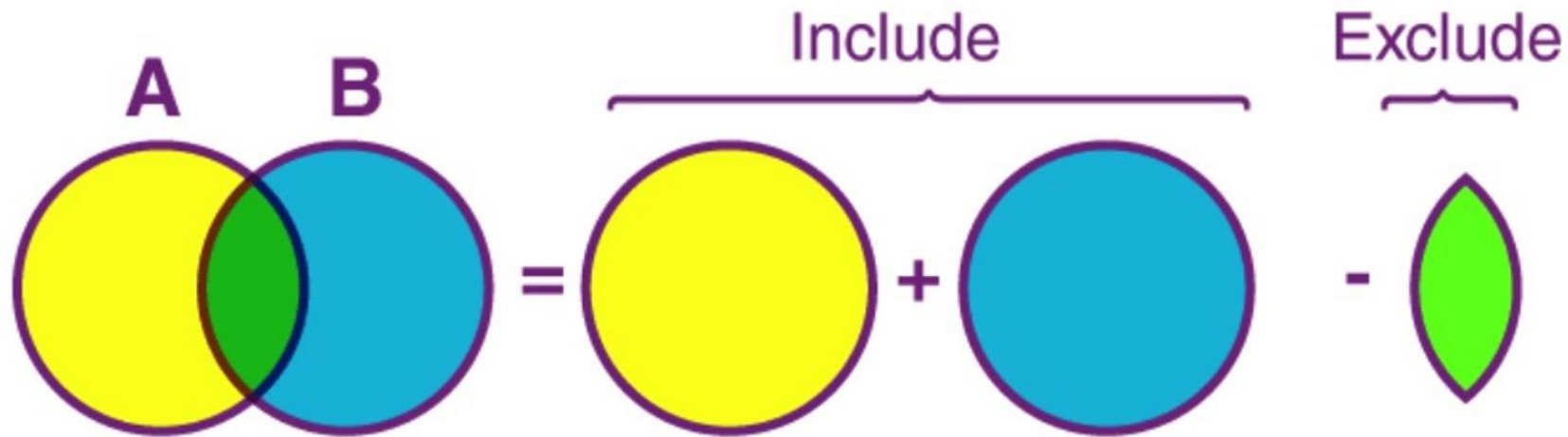
By event relationship, say either A or B happen



How to calculate $P(A \text{ or } B)$, namely $P(A \cup B)$?

Calculation of Event Probability?

Addition Rules

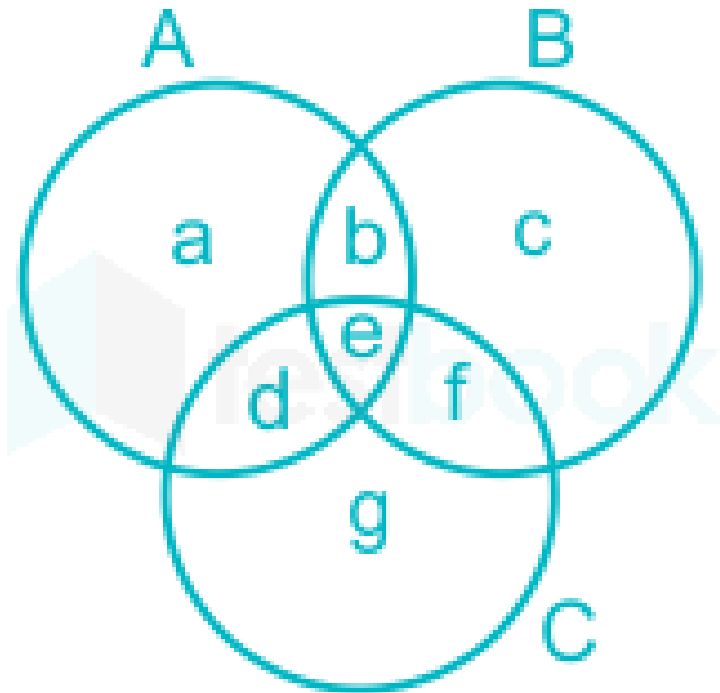


$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Addition Rules for more events?

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

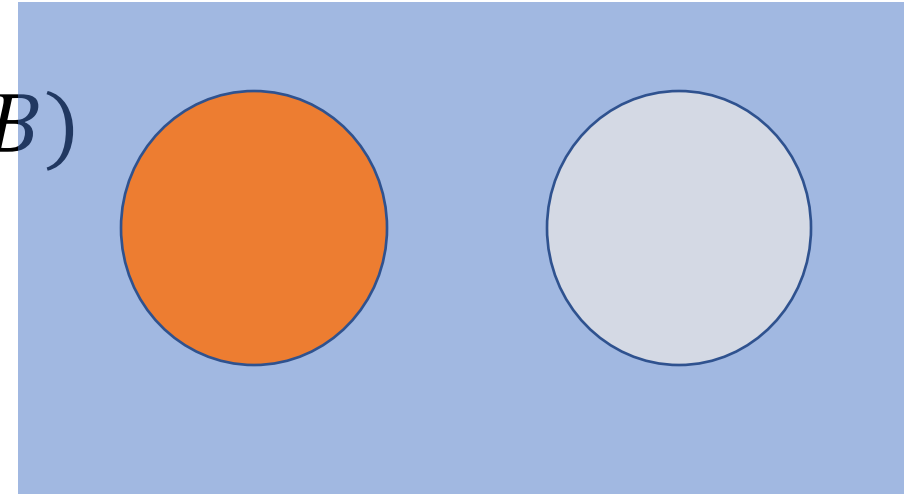


Focus on two events.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

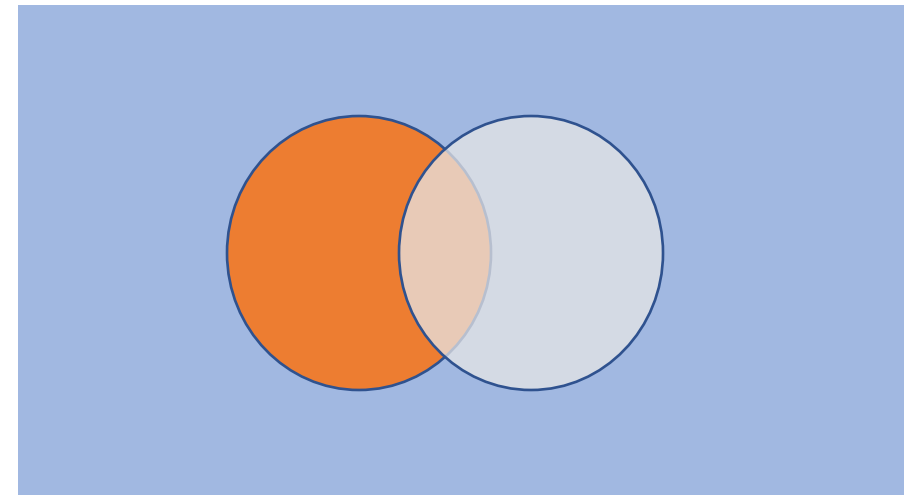
- Case: $P(A \cap B) = 0$

- If A happens, B cannot happen



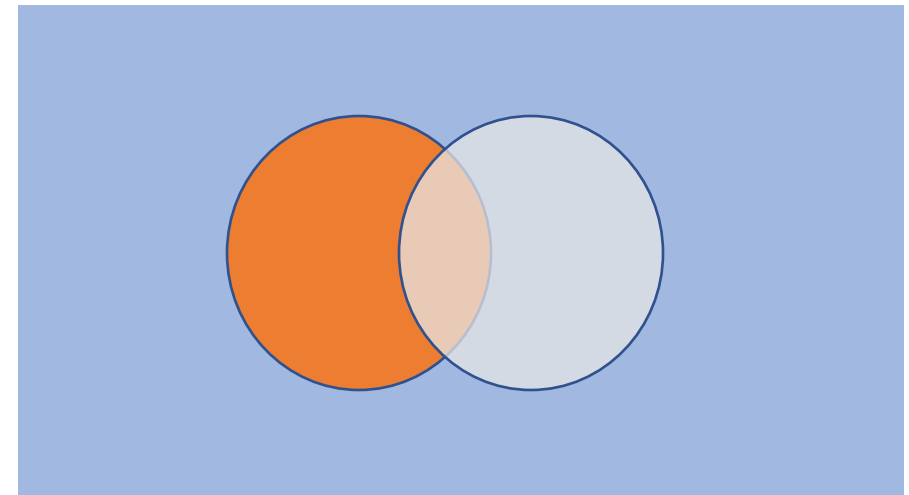
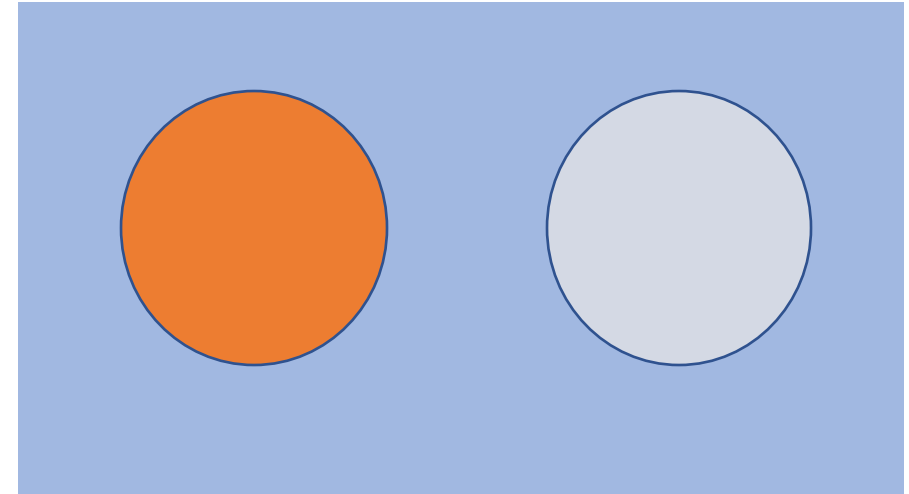
- Case: $P(A \cap B) \neq 0$

- If A happens, B may also happen



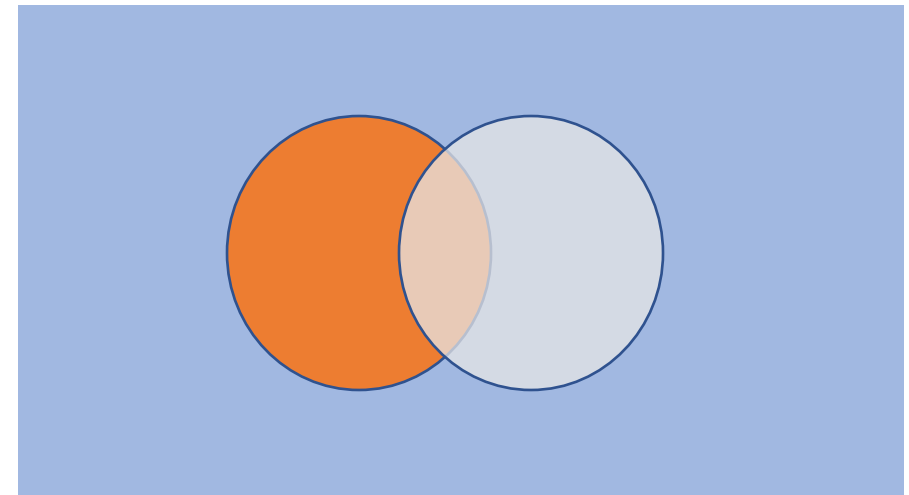
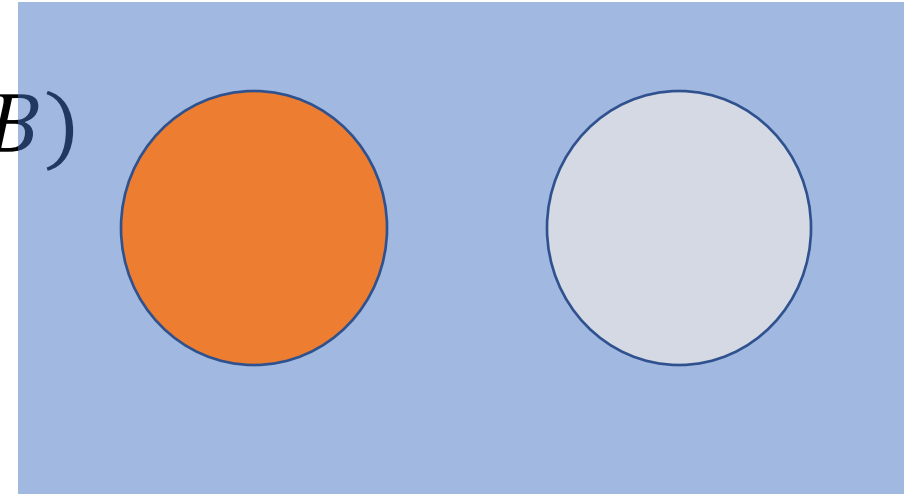
Focus on two events.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 - Case: $P(A \cap B) = 0$
 - If A happens, B cannot happen
 - A and B are disjoint (mutually exclusive)
-
- Case: $P(A \cap B) \neq 0$
 - If A happens, B may also happen
 - A and B are joint (not mutually exclusive)



Focus on two events.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If A and B are disjoint
- $P(A \cup B) = P(A) + P(B)$
- If A and B are joint
- $P(A \cup B) < P(A) + P(B)$



Example 8

- Toss a coin **2** times:
 $\{TT, HT, TH, HH\}$ equally likely

1 \ 2	T	H
T	TT	TH
H	HT	HH

- $H1$: the first outcome is H
- $T1$: the first outcome is T
- $H2$: the second outcome is H

Event	H1	T1	H1 or T1	H1 and T1
Probability	0.5	0.5	1	0

Event	H1	H2	H1 or H2	H1 and H2
Probability	0.5	0.5	0.75	0.25

Example 8

Event	H1	T1	H1 or T1	H1 and T1
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Event	H1	H2	H1 or H2	H1 and H2
Probability	0.5	0.5	0.75	0.25

- $H1$: the first outcome is H
- $T1$: the first outcome is T
- $H2$: the second outcome is H

$$P(\textcolor{red}{H1} \cup \textcolor{blue}{T1}) = P(H1) + P(T1)$$

mutually exclusive

Example 8

Event	H1	T1	H1 or T1	H1 and T1
Probability	0.5	0.5	1	0

Event	H1	H2	H1 or H2	H1 and H2
Probability	0.5	0.5	0.75	0.25

- $H1$: the first outcome is H
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- $H2$: the second outcome is H

$$P(\textcolor{red}{H1} \cup \textcolor{blue}{T1}) = P(H1) + P(T1)$$

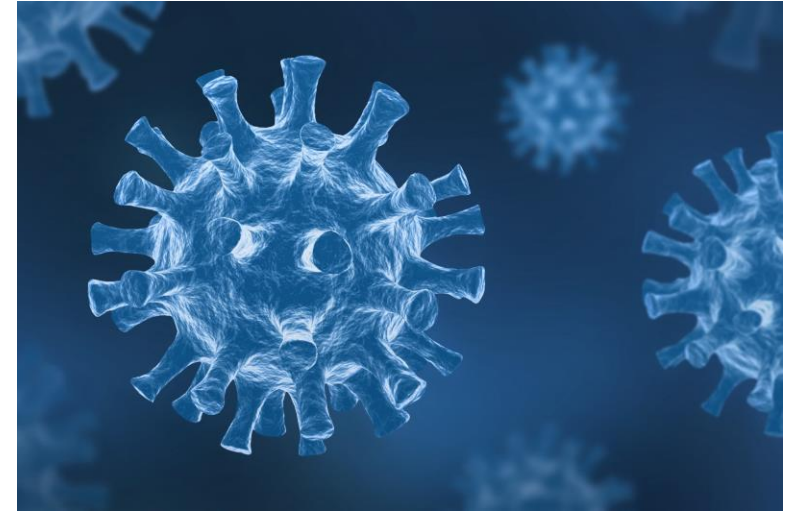
$$P(\textcolor{red}{H1} \cup \textcolor{red}{H2}) < P(H1) + P(H2)$$

not mutually exclusive

Exercise: Joint or disjoint

A: Infected with COVID-19

B: Nucleic acid test is negative



Experiment: toss a coin **3** times.

A: “exactly **2** heads”

B: “exactly **2** tails”

